

ARTICLES

Acceptor-induced reorientation enables vertically and laterally optimized charge-transport devices with a single polymer

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Abstract Molecular orientation dictates anisotropic charge transport in conjugated polymers. Here, we show that acceptor-induced reorientation enables a polymer system to align with both lateral and vertical transport geometries. A series of terpolymers (PTTzF-Th, PTTzF-TT, and PTTzF-BT) was synthesized by partially replacing 2F-thiophene units in PTTzF with non-fluorinated thiophene comonomers. All terpolymers display edge-on orientation in neat films, with PTTzF-BT having the highest edge-on fraction; upon blending with Y6, acceptor templating induces a pronounced reorientation toward face-on, most strongly in PTTzF-BT:Y6. As a result, PTTzF-BT achieves the highest thermoelectric (conductivity: 3.7 S cm^{-1} ; Seebeck coefficient: $69.3 \text{ } \mu\text{V K}^{-1}$) and photovoltaic (power conversion efficiency: 16.1%) performance among the series. Mechanistic analysis highlights coupled kinetic and thermodynamic effects: in neat films, a longer film-formation window favors edge-on orientation; in blends, improved donor-acceptor miscibility and slower acceptor crystallization enhance templating and promote reorientation to face-on. These insights demonstrate a conceptual framework for programmable molecular orientation in conjugated polymers and illustrate how a single polymer platform can be aligned with distinct charge transport geometries in different organic electronic devices.

Keywords conjugated polymers, molecular orientation, organic solar cells, organic thermoelectrics

1 Introduction

The molecular orientation of conjugated polymers governs charge transport anisotropy and plays a pivotal role in determining device performance. In edge-on orientation, the π - π stacking direction lies parallel to the substrate, favoring in-plane charge transport that benefits organic thermoelectrics devices (OTEs) [1,2]. Conversely, in face-on orientation, π - π stacking is oriented perpendicular to the substrate, enabling efficient vertical charge extraction essential for organic solar cells (OSCs) [3]. Aligning molecular orientation with device-specific charge transport directions offers a robust strategy for optimizing organic electronic performance [4–6].

The advent of non-fullerene acceptors, exemplified by Y6, has revitalized interest in semicrystalline polythiophene donors for OSCs, owing to their scalable synthesis, low cost, and tunable crystallinity [7,8]. Notably, many representative polymers—such as PDCBT [9] and PTVT-BT [10] from Hou's group, P5TCN-2F [11] and P5TCN-F25 [12] from Duan's group, and PT-4F from Kim's group [13]—exhibit predominantly edge-on molecular orientation in neat films, while their corresponding blends with acceptors tend to adopt a face-on orientation. In our previous study on the effect of incorporating a fluorine-containing third component on structural ordering in PTTz-based polymers, we observed the same phenomenon: although an increasing ratio of the third component promoted edge-on orientation in neat films, the corresponding blends with acceptors still achieved stable face-on orientation [14]. Despite the frequent observation of this acceptor-induced reorientation, its underlying mechanism and implications for device

performance remain insufficiently understood.

Acceptor-induced reorientation presents both opportunities and challenges for morphology control in conjugated polymers. On the one hand, the acceptor's templating effect enables programmable molecular orientation, allowing a single polymer to adopt edge-on alignment in OTEs and face-on alignment in OSCs. On the other hand, orientation evolution during solution processing is governed by multiple interdependent factors, including polymer aggregation, film-formation kinetics, and donor-acceptor miscibility [15–17]. These factors are intricately coupled, and deciphering their individual contributions remains a significant challenge.

To address this, we synthesized a PTTzF-derived terpolymer series (PTTzF-Th, PTTzF-TT, PTTzF-BT) by partially substituting 2F-thiophene units in the parent polymer with non-fluorinated thiophene comonomers. All three terpolymers adopted predominant edge-on orientation in neat films and underwent varying degrees of face-on reorientation upon blending with Y6. Among them, PTTzF-BT exhibited the most favorable orientation profile, combining the highest edge-on fraction in neat films with the largest face-on shift in blends, which aligned well with the charge transport requirements of OTE and OSC devices. This orientation adaptability translated into superior multifunctional performance, delivering the highest conductivity (3.7 S cm^{-1}) and Seebeck coefficient ($69.3 \text{ } \mu\text{V K}^{-1}$) in OTEs, along with a power conversion efficiency (PCE) of 16.1% in OSCs. By correlating orientation with aggregation behavior, film-formation kinetics, and donor-acceptor miscibility, we elucidated the coupled kinetic and thermodynamic influences on orientation evolution and formulated design principles for morphol-

ogy-programmable conjugated polymers within this class of PTTzF-based systems.

2 Results and discussion

Figure 1a illustrates the molecular structures of the terpolymers (PTTzF-Th, PTTzF-TT, and PTTzF-BT), in which 20 mol% of thiophene (Th), thienothiophene (TT), and bithiophene (BT) units were incorporated into the PTTzF backbone via Pd-catalyzed Stille coupling, respectively [18,19]. For a fair comparison, the parent polymer PTTzF was also synthesized under identical conditions. Detailed synthetic routes and experimental procedures are provided in the Supporting Information online (Scheme S1, Figures S1–S7). Gel permeation chromatography (GPC) measurements revealed comparable number-average molecular weights (M_n) in the range of 30–36 kDa for all polymers, ensuring that the subsequent analysis reflects structural variations rather than molecular-weight differences.

Cyclic voltammetry (CV) measurements of the polymer films were conducted to determine their energy levels (Figure S8). Relative to the parent polymer PTTzF (−5.66/−3.74 eV), the terpolymers exhibited slightly upshifted highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) levels, measured at −5.64/−3.73 eV (PTTzF-Th), −5.61/−3.69 eV

(PTTzF-BT), and −5.59/−3.68 eV (PTTzF-TT), reflecting their somewhat enhanced donor nature (Figure 1b). The moderate HOMO upshift is beneficial in two ways: it facilitates hole generation during p-type doping [20,21] and improves energy alignment with acceptors, thereby promoting exciton dissociation and overall device performance [22–24].

Density functional theory (DFT, B3LYP/6-31G(d,p)) calculations indicate that PTTzF adopts a nearly planar backbone conformation, stabilized by intrachain noncovalent interactions such as S⋯N and F⋯S (Figure 1c) [25]. These conformational locks extend π -conjugation and promote compact molecular packing [26–28]. In contrast, the terpolymer models—most prominently PTTzF-BT—exhibited increased backbone torsion ($\phi > 9.1^\circ$) owing to reduced intrachain noncovalent stabilization. Such conformational relaxation leads to enhanced solution processability, as evidenced by markedly improved solubility in chloroform (Figure S9), consistent with reduced interchain aggregation [29–32].

While UV-vis absorption spectra of all polymers appeared similar in solution and film, closer inspection, as shown in Figure 1d, e, revealed distinct vibronic features, indicating subtle variations in their aggregation and electronic coupling behaviors. To elucidate the different aggregation behaviors, we first quantitatively analyzed the intensity ratio between the 0-0 and 0-1 transitions (I_{0-0}/I_{0-1}) in the polymer films. Relative to PTTzF, the terpolymers exhibited

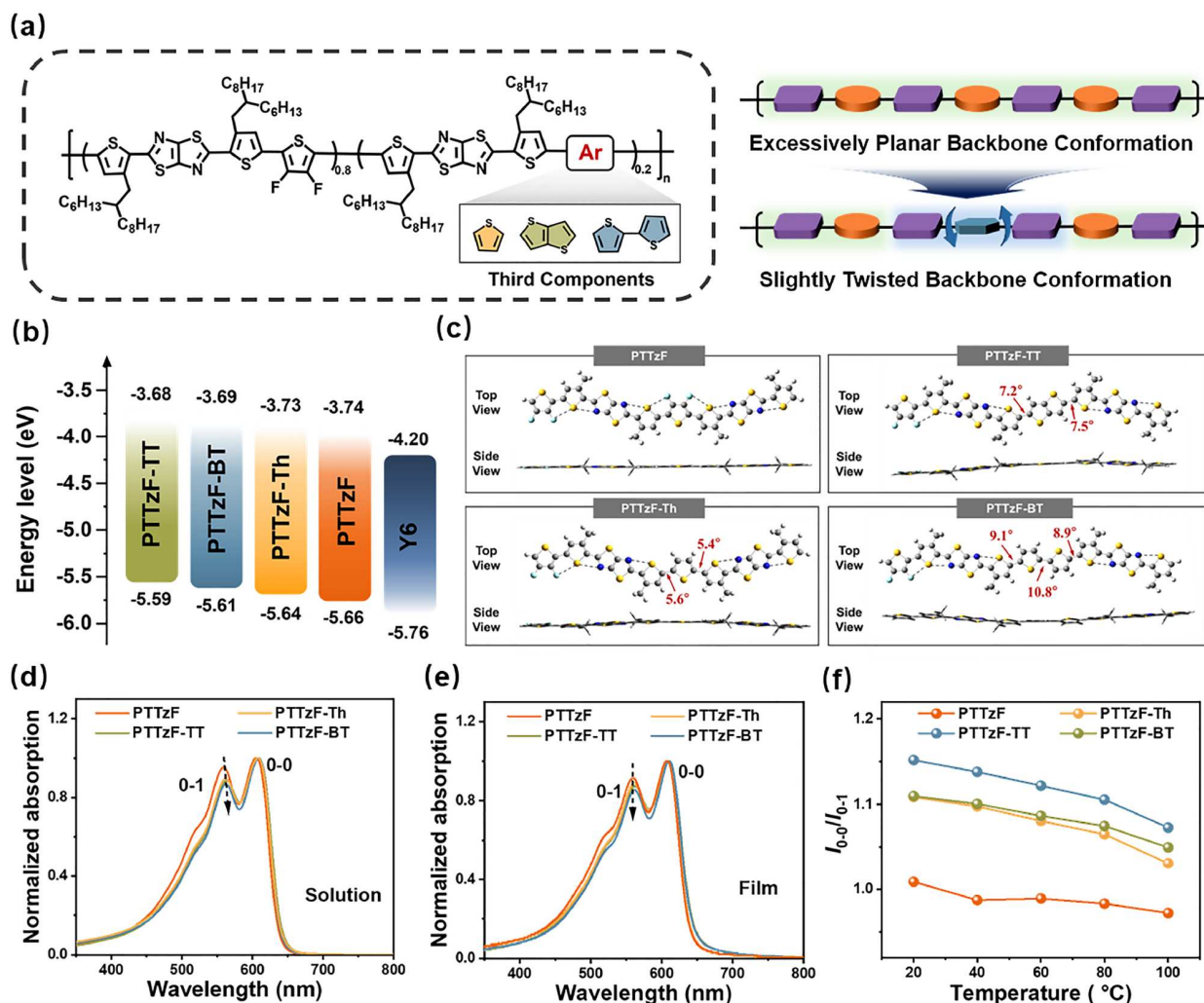


Figure 1 (Color online) (a) Chemical structures and illustration of the backbone conformations of polymers. (b) Energy level diagrams of materials used in this study. (c) Optimized molecular conformations of polymers by DFT calculations. The UV-vis absorption spectra of the polymer in (d) solution and (e) film state. (f) Plot of the I_{0-0}/I_{0-1} versus temperature of polymers.

higher I_{0-0}/I_{0-1} ratios, with PTTzF-BT showing the highest value of 1.176, indicative of strengthened J-aggregate characteristics arising from increased backbone torsion [33,34]. This aggregation tendency lowers the exciton binding energy and enhances exciton dissociation, promoting efficient photoelectric conversion.

In addition, we conducted temperature-dependent UV-vis absorption measurements to probe the thermal evolution of aggregation (Figure S10). As the films were heated from 20 to 100 °C, the 0-0 band intensity decreased for all the polymers; the ratio $I_{\max}^{100^\circ\text{C}}/I_{\max}^{20^\circ\text{C}}$ is smaller for the terpolymers, with values of 0.88 for PTTzF-Th, 0.91 for PTTzF-TT, and 0.89 for PTTzF-BT, compared with 0.94 for PTTzF (Figure 1f and Table S1, Supporting Information online). The more pronounced reduction of the 0-0 band intensity with increasing temperature in the terpolymers not only supports their enhanced temperature-dependent aggregation property, beneficial for morphology formation and processing, but also suggests a weakening of the overall aggregation strength relative to PTTzF [35].

To elucidate how backbone structure influences molecular orientation, we conducted grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements on both neat polymer films and their corresponding Y6 blends. The 2D GIWAXS patterns are seen from Figure 2a, b, and the corresponding molecular stacking parameters are summarized in Tables S2 and S3. In neat films, PTTzF exhibited a predominantly face-on orientation, as evidenced by the pronounced (010) π - π stacking peak in the out-of-plane (OOP) direction, whereas all three terpolymers displayed a clear edge-on-dominated orientation, with the (010) π - π stacking peak

primarily located in the in-plane (IP) direction. To quantify the orientation distributions, we performed azimuthal angle integration of the (010) diffraction intensity (Figure 2c), followed by deconvolution into face-on and edge-on contributions. The edge-on proportion in neat polymer films ($P_{\text{edge-on}}^{\text{neat}}$) increased from 42% in PTTzF to 75% in PTTzF-Th, 73% in PTTzF-TT, and 83% in PTTzF-BT. This trend indicates that partial replacement of 2F-thiophene units with more flexible, non-fluorinated thiophene comonomers disrupts backbone planarity and enhances in-plane stacking propensity, thereby favoring edge-on molecular orientation.

Upon blending with Y6, all the blend films exhibit a pronounced reorientation of the (010) π - π stacking peak toward the face-on direction. Azimuthal integration of the (010) scattering yielded the apparent face-on proportion ($P_{\text{face-on}}^{\text{blend}}$) of 82% for PTTzF:Y6, 76% for PTTzF-Th:Y6, 92% for PTTzF-TT:Y6, and 91% for PTTzF-BT:Y6 (Figure 2d, e). To quantify this reorientation, we here defined an orientation increment (OI) as $\text{OI} = P_{\text{face-on}}^{\text{blend}} - P_{\text{face-on}}^{\text{neat}}$. The calculated OI values are 24% (PTTzF), 51% (PTTzF-Th), 65% (PTTzF-TT), and 73% (PTTzF-BT), with PTTzF-BT showing the largest reorientation. These results demonstrate that Y6 acts as an effective template, inducing the π - π stacking network to orient along the film normal and thereby forming a morphology highly conducive to vertical charge transport. Consistent with this picture, GIWAXS of a neat Y6 film reveals a predominantly face-on orientation (Figure S11). It should also be noted that the extent of such acceptor-induced reorientation varies markedly among the polymers possessing

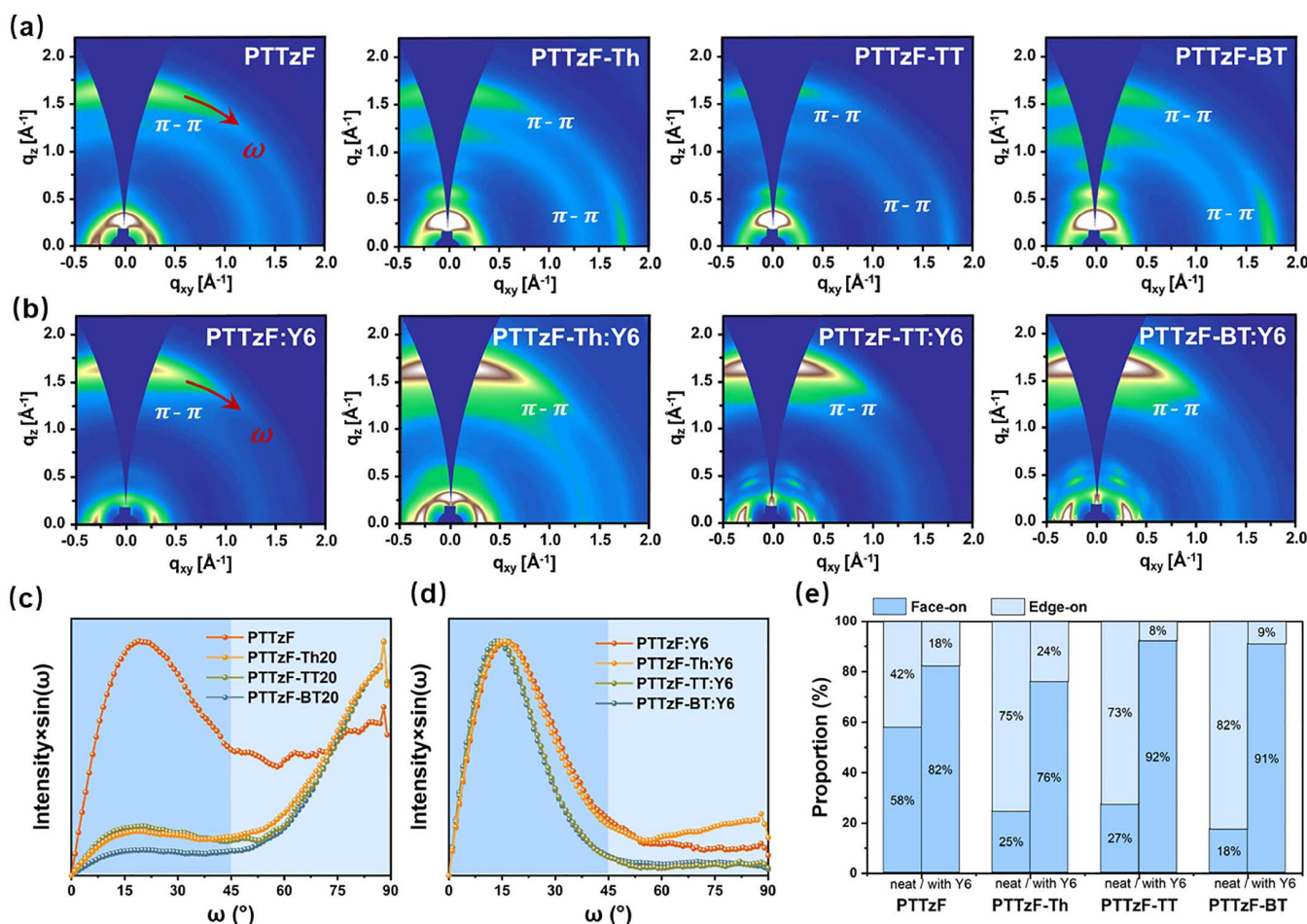


Figure 2 (Color online) 2D GIWAXS patterns of the (a) neat polymer films and (b) polymer:Y6 blend films. The azimuthal profiles from (010) π - π peak in (c) neat polymer films and (d) polymer:Y6 blend films. (e) Proportions of face-on and edge-on orientation.

different aggregation characteristics.

To evaluate edge-on-dominated films for OTEs (Figure 3a), FeCl₃-doped neat films showed conductivities (σ) of 0.54, 1.29, 2.02, and 3.70 S cm⁻¹ for PTTzF, PTTzF-Th, PTTzF-TT, and PTTzF-BT, respectively. The higher σ of PTTzF-TT relative to PTTzF and PTTzF-Th is consistent with its moderately upshifted HOMO level, which facilitates p-type doping. PTTzF-BT exhibits a further increase in σ , which may be associated with its higher HOMO level and the edge-on orientation observed in the pristine film, both of which could contribute to more efficient lateral charge transport upon doping. The corresponding Seebeck coefficients (S) were 38.9, 39.8, 50.44, and 69.3 $\mu\text{V K}^{-1}$ for PTTzF, PTTzF-Th, PTTzF-TT, and PTTzF-BT, respectively (Figure 3b). The resulting power factors ($S^2\sigma$) increased in the order of PTTzF, PTTzF-Th, PTTzF-TT, and PTTzF-BT (Figure 3c) [36]. These results suggest that the terpolymer strategy improves thermoelectric performance by enhancing p-type doping efficiency through higher HOMO energy levels and by promoting in-plane charge transport, as supported by the edge-on molecular orientation observed in pristine films.

To evaluate the photovoltaic performance of the face-on-dominated polymer:Y6 blend films formed via acceptor-induced reorientation, OSCs with an ITO/PEDOT:PSS/polymer:Y6/H75/Ag architecture were also fabricated and systematically characterized (Figure 3d) [37–39]. The J - V curves and statistical distributions of

PCEs are shown in Figure 3e, f, and the corresponding photovoltaic parameters are summarized in Table S4. The PTTzF:Y6-based device showed a PCE of 13.80% with a short-circuit current (J_{SC}) of 22.85 mA cm⁻², an open-circuit voltage (V_{OC}) of 0.851 V, and a fill factor (FF) of 70.93%. The terpolymer:Y6-based devices displayed reduced V_{OC} values due to the upshifted HOMO levels of the terpolymers; however, their substantially enhanced J_{SC} values resulted in overall higher PCEs, following the order of 14.73% (PTTzF-Th:Y6), 15.96% (PTTzF-TT:Y6), and 16.07% (PTTzF-BT:Y6). The integrated current densities calculated from the external quantum efficiency (EQE) spectra agreed well with those obtained from the J - V characteristics, exhibiting negligible discrepancies. Moreover, the EQE spectra confirmed a markedly stronger photoresponse across the entire absorption region for the terpolymer:Y6-based devices compared to the PTTzF:Y6-based counterpart (Figure 3g).

To gain deeper insight into the origin of the enhanced J_{SC} values, we analyzed the photocurrent density (J_{ph}) as a function of the effective voltage (V_{eff}). As depicted in Figure 3h, the terpolymer:Y6-based devices exhibited an exciton dissociation probability of approximately 93% and a charge-collection efficiency exceeding 81%, both higher than those of the PTTzF:Y6 counterpart (88% and 76%, respectively). These results validate that the terpolymer:Y6-based devices facilitate more efficient exciton separation and

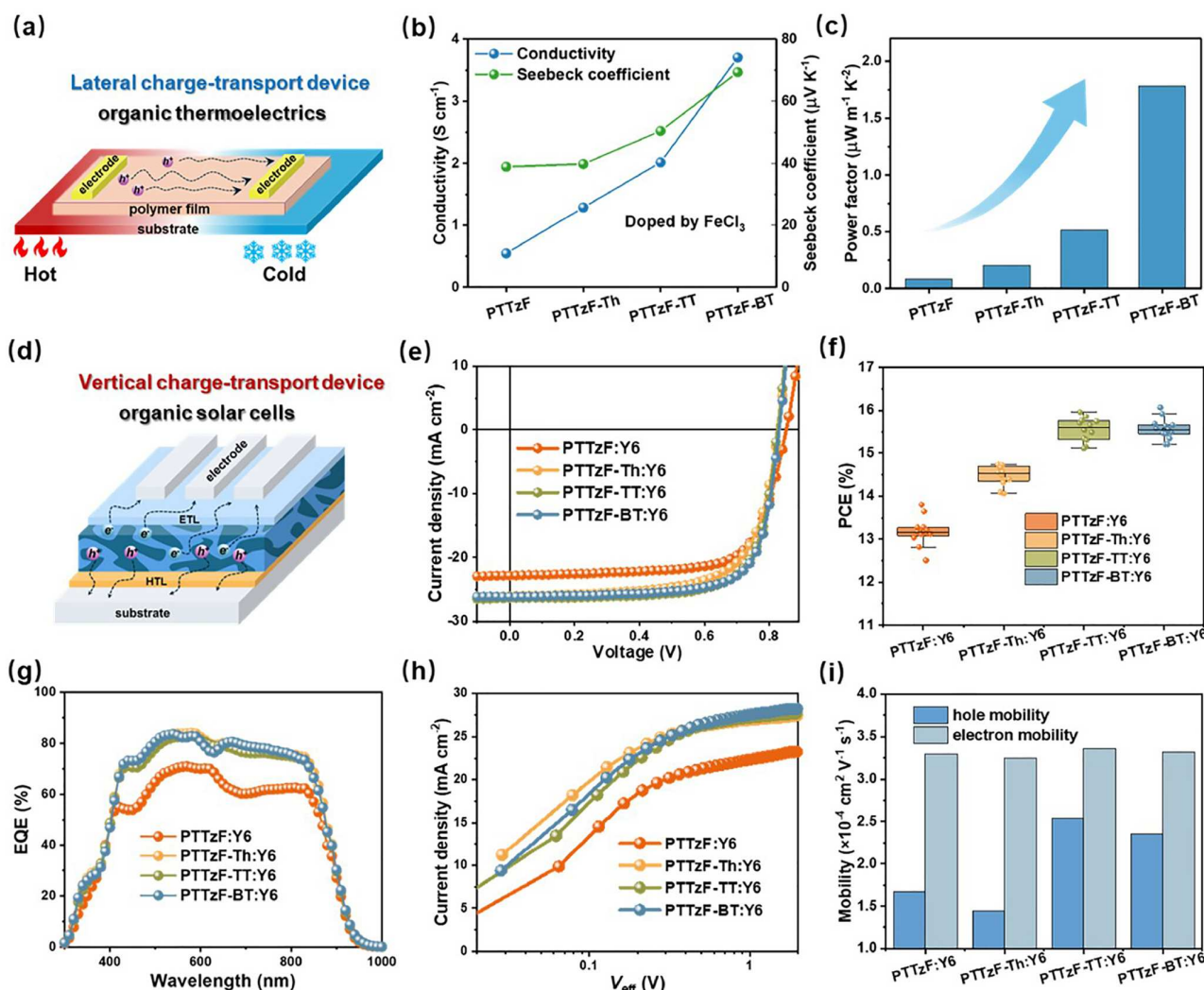


Figure 3 (Color online) (a) Illustration of the organic thermoelectric device. (b) Plot of the conductivity and Seebeck coefficient of FeCl₃-doped polymers. (c) Power factor of FeCl₃-doped polymers. (d) Illustration of the organic solar cell device. (e) J - V curves and (f) statistical distribution of PCEs with 12 individual devices. (g) Corresponding EQE spectra, (h) photocurrent versus effective voltage, and (i) hole/electron mobilities of four polymer-based devices.

carrier extraction than the PTTzF:Y6-based ones.

Space-charge-limited current (SCLC) measurements showed hole mobilities of 1.67×10^{-4} (PTTzF:Y6), 1.45×10^{-4} (PTTzF-Th:Y6), 2.54×10^{-4} (PTTzF-TT:Y6), and $2.36 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ (PTTzF-BT:Y6). The electron mobilities were comparable across all the blends, measured at approximately $3.3 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ (Figure 3i and Figure S12), resulting in the PTTzF-TT:Y6 and PTTzF-BT:Y6 systems exhibiting well-balanced carrier transport, which favors efficient charge extraction and mitigates space-charge effects [40]. The higher hole mobilities in these blends originate from efficient vertical π - π coupling pathways through their larger face-on fractions [41]. Such advantageous charge-transport characteristics account for the superior PCEs achieved by the PTTzF-TT:Y6- and PTTzF-BT:Y6-based devices. Notably, PTTzF-BT stands out by delivering the best overall performance in both OTE and OSC applications.

To investigate how film-formation kinetics govern molecular orientation, we conducted *in situ* UV-vis absorption monitoring during the spin-casting of both neat polymer and blend films [42,43]. For the neat polymer films, the time required to reach steady-state absorption (τ_{polymer}) during spin coating was determined to be 545 ms for PTTzF, 697 ms for PTTzF-TT, 731 ms for PTTzF-Th, and 818 ms for PTTzF-BT (Figure S13). We interpret this result as indicating that the backbone torsion and microstructural disorder introduced by terpolymerization delay orientation freezing, providing greater opportunity for segmental rearrangement at the substrate, favoring edge-on orientation with dominant lateral π - π pathways.

In the blend films, the temporal evolution of the acceptor absorption was tracked through the continuous redshift of the Y6

main band, which reflects a transition from dispersed to semicrystalline states (Figure 4a, b). The redshift durations (τ_{Y6}) were determined to be 103 ms for PTTzF:Y6, 178 ms for PTTzF-Th:Y6, 296 ms for PTTzF-TT:Y6, and 255 ms for PTTzF-BT:Y6. Larger τ_{Y6} values correspond to slower crystallization of the acceptor, allowing greater interfacial rearrangement and donor-acceptor registry [44,45]. We note that markedly prolonged τ_{Y6} values appeared in terpolymer-based blends, particularly for PTTzF-TT:Y6 and PTTzF-BT:Y6, which possess higher face-on fractions. Overall, the irregular structural composition of terpolymers extends the film-formation kinetics, promoting the development of edge-on orientation in their neat films, yet conversely reorienting the molecular packing toward a face-on configuration along the film normal in the corresponding blend films with the acceptor.

To evaluate the interfacial miscibility between the donor polymers and Y6, we employed the Wu model in combination with contact-angle measurements using water (H_2O) and diiodomethane (CH_2I_2 , DIM) as probe liquids (Figure 4c). From the dispersive and polar components of the surface free energy, the total surface energy (γ) was obtained, and the Flory-Huggins interaction parameter (χ) with Y6 was subsequently estimated (Figure 4d, Table S5) [46,47]. The values are: PTTzF, $\gamma = 31.25 \text{ mN m}^{-1}$ and $\chi = 0.95 \text{ K}$; PTTzF-Th, $\gamma = 32.05 \text{ mN m}^{-1}$ and $\chi = 0.83 \text{ K}$; PTTzF-TT, $\gamma = 32.17 \text{ mN m}^{-1}$ and $\chi = 0.81 \text{ K}$; PTTzF-BT, $\gamma = 32.81 \text{ mN m}^{-1}$ and $\chi = 0.72 \text{ K}$. Compared with PTTzF, all the terpolymers exhibited higher γ and lower χ , reflecting improved thermodynamic miscibility with Y6. Among them, PTTzF-BT showed the smallest χ , suggesting the most favorable donor-acceptor mixing and a pronounced templating effect of Y6 crystallization that drives reorientation in the blend.

This study simultaneously involves a broad set of parameters

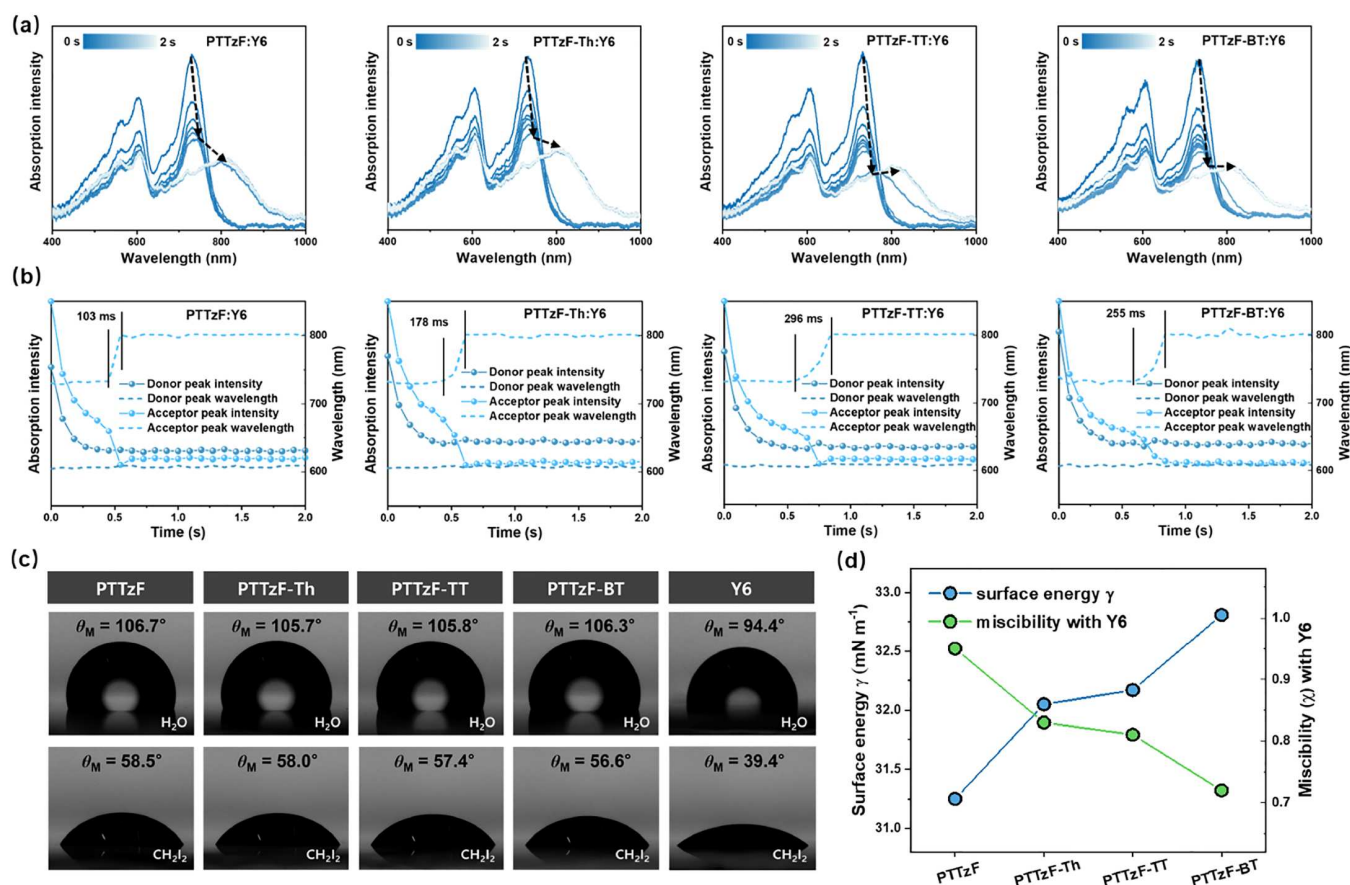


Figure 4 (Color online) *In-situ* UV-vis absorption (a) line profile of polymer:Y6 blends and (b) corresponding time evolution of peak position and intensity. (c) Contact angles of water and diiodomethane on neat films. (d) Plot of the surface energy and Flory-Huggins parameter χ of polymers with Y6.

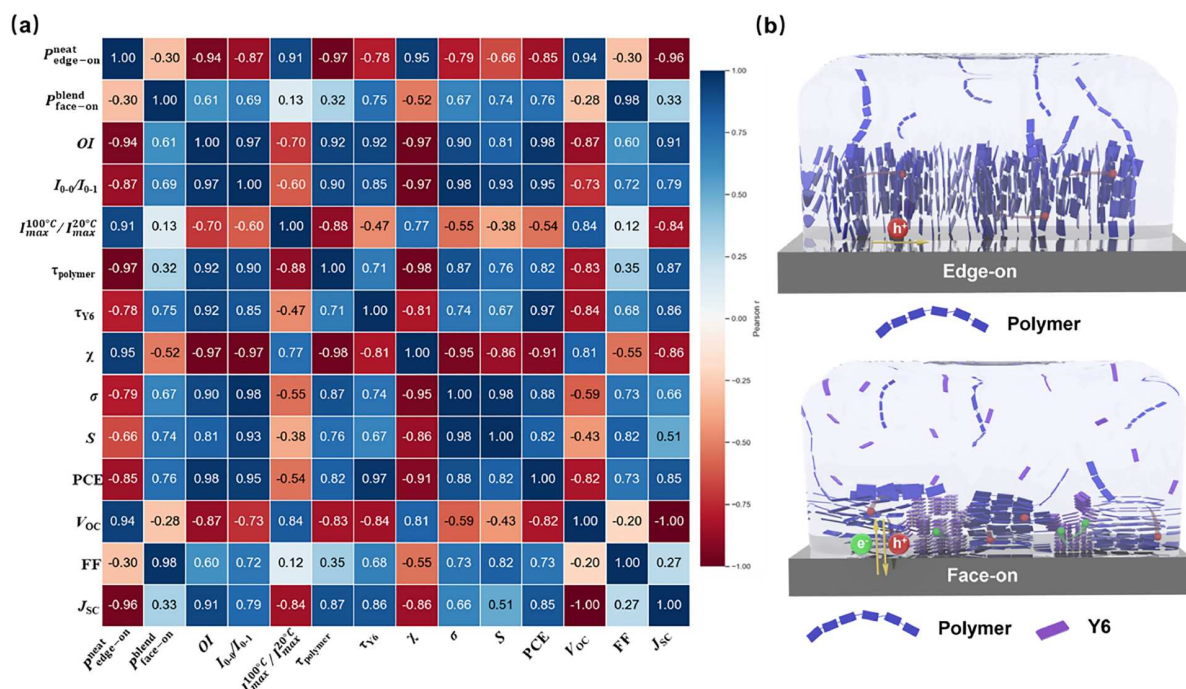


Figure 5 (Color online) (a) Pearson correlation matrix of key parameters. (b) Illustration of orientation and direction of charge transport of neat polymer film and polymer:Y6 blend film.

across multiple dimensions, including orientation and reorganization descriptors ($p_{\text{edge-on}}^{\text{neat}}$, $p_{\text{face-on}}^{\text{blend}}$, OI), spectroscopic indicators (I_{0-0}/I_{0-1} , $I_{\text{max}}^{100^\circ\text{C}}/I_{\text{max}}^{20^\circ\text{C}}$), film-formation kinetics (τ_{polymer} , τ_{Y6}), thermodynamic interaction (χ), and device performance metrics for both thermoelectric and photovoltaic functions (σ , S , and PCE with V_{OC} , FF , and J_{SC}). To elucidate interrelationships among variables, we computed Pearson correlation coefficients and summarized them in a correlation matrix (Figure 5a). Pearson's r ranges from -1 to $+1$; the sign indicates direction (positive/negative), and the magnitude reflects strength, which we interpret as $|r| < 0.3$ weak, 0.3 – 0.7 moderate, and >0.7 strong [48]. This visualization highlights global trends and parameter couplings.

Examination of the correlation matrix reveals two clear origins of molecular orientation in neat polymer and blend films. For neat polymer films, the edge-on fraction appears to be closely associated with film-formation kinetics. A positive correlation is observed between the edge-on fraction and the drying timescale τ_{polymer} ($r \approx 0.97$), suggesting that slower solvent evaporation provides a longer temporal window for rearrangement of polymer chains, thereby favoring edge-on orientation. In contrast, the reorientation behavior in blends is linked to both thermodynamic miscibility and acceptor crystallization kinetics. The orientation increment OI shows a negative correlation with the Flory-Huggins parameter χ ($r \approx -0.97$), indicating that improved donor–acceptor miscibility is accompanied by more substantial orientation reorganization. At the same time, OI correlates positively with the τ_{Y6} ($r \approx 0.92$), consistent with the notion that slower nucleation and growth of the acceptor afford extended time for interfacial rearrangement and promote a more complete reorientation of the π - π stacking network. Ultimately, the orientation reorganization induced by the terpolymer strategy directly translates into device performance. Enhanced edge-on orientation in neat polymer films improves electrical conductivity without sacrificing the S in OTEs, while a higher face-on fraction in blends supports more efficient vertical charge transport and leads to increased J_{SC} and PCE values in OSCs (Figure 5b).

3 Conclusions

In summary, we clarified the mechanistic basis of acceptor-induced reorientation, wherein the interplay between film-formation kinetics and donor–acceptor mixing thermodynamics jointly governed the final molecular orientation. Introducing non-fluorinated thiophene comonomers into the PTTzF backbone increased torsion and extended the drying window prior to orientation freezing, steering neat films toward an edge-on orientation with continuous in-plane π - π pathways. In Y6 blends, improved donor–acceptor miscibility and retarded acceptor crystallization strengthened acceptor templating, guiding reorientation of π - π stacking along the film normal to a face-on geometry. Among the PTTzF series, the PTTzF-BT most effectively harmonized the lateral and vertical charge-transport properties, leading to substantial improvements in both OTEs and OSCs. These findings establish PTTzF-BT as a model for programmable orientation in conjugated polymers and highlight how orientation engineering can be leveraged as a unifying handle for optimizing diverse organic electronic functions.

Conflict of interest

The authors declare no conflict of interest.

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Supporting information

The supporting information is available online at <http://chem.scichina.com> and <http://link.springer.com/journal/11426>. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

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